

Solving quantum mechanical problems with VQE algorithm

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Plans for the week of March 10-14: Solving quantum mechanical problems with VQE algorithm

- 1 Discussion of our final model, the Lipkin model, a two-qubit and a four-qubit system, reminder from last week, code examples and theory
- 2 Jordan-Wigner transformation
- 3 Applications to other Hamiltonians, pairing Hamiltonian (codes and theory)
- 4 If we get time: Start discussion of Fourier transforms, continuous and discrete,, general motivation
- 5 Discussions and work on project 1 during lab sessions
- 6 Video of lecture
- 7 Whiteboard notes

Hamiltonian in terms of Pauli strings

We have seen during the previous weeks how to rewrite many-body Hamiltonians in terms of Pauli strings built from

$$I, \quad X, \quad Y, \quad Z,$$

so that they are ready for quantum-computing implementations.

- the Lipkin Hamiltonian in quasispin form,
- detailed mappings to Pauli operators,
- compressed and direct qubit encodings,
- second-quantized fermionic operators $a^\dagger, a$,
- the nuclear pairing Hamiltonian as a second worked example,
- a fuller Jordan–Wigner mapping from fermions to qubits.

Notation

We use:

- $a_p^\dagger$: fermion creation operator in orbital p ,
- a_p : fermion annihilation operator in orbital p ,
- $n_p = a_p^\dagger a_p$: number operator,
- X_i, Y_i, Z_i : Pauli operators on qubit i ,
- I_i : identity on qubit i .

Throughout, we use only

$$a_p^\dagger, \quad a_p.$$

For quasispin systems we use collective operators

$$J_x, J_y, J_z, \quad J_\pm = J_x \pm iJ_y.$$

For pairing we use local quasispin operators

$$S_p^\pm, \quad S_p^z.$$

Why Pauli decompositions matter

Quantum algorithms typically require a Hamiltonian of the form

$$H = \sum_{\alpha} h_{\alpha} P_{\alpha}, \quad P_{\alpha} \in \{I, X, Y, Z\}^{\otimes n}.$$

Once a Hamiltonian is written in this way, one can:

- estimate energies in VQE,
- generate Trotterized time evolution,
- build QAOA circuits,
- exploit commutativity and measurement grouping.

So the main preprocessing step is:

rewrite the many-body Hamiltonian as a weighted sum of Pauli strings.

Lipkin Hamiltonian

We consider the Lipkin Hamiltonian in quasispin representation

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2.$$

Here:

- N is the number of particles,
- ϵ sets the one-body splitting,
- V controls pair scattering between the two levels,
- W weights the quadratic J_z^2 term.

Quasispin algebra

The collective operators satisfy the $su(2)$ algebra

$$[J_x, J_y] = iJ_z, \quad [J_y, J_z] = iJ_x, \quad [J_z, J_x] = iJ_y,$$

or, equivalently,

$$[J_z, J_{\pm}] = \pm J_{\pm}, \quad [J_+, J_-] = 2J_z.$$

The basis states are

$$|j, m\rangle, \quad m = -j, -j+1, \dots, j,$$

with

$$J_z |j, m\rangle = m |j, m\rangle.$$

Action of ladder operators

The ladder operators act as

$$J_+ |j, m\rangle = \sqrt{j(j+1) - m(m+1)} |j, m+1\rangle,$$

$$J_- |j, m\rangle = \sqrt{j(j+1) - m(m-1)} |j, m-1\rangle.$$

Therefore

$$J_+^2 |j, m\rangle = \sqrt{j(j+1) - m(m+1)} \sqrt{j(j+1) - (m+1)(m+2)} |j, m+2\rangle,$$

and analogously for J_-^2 .

Useful identity for the interaction term

Using

$$J_{\pm} = J_x \pm iJ_y,$$

we compute

$$J_+^2 + J_-^2 = (J_x + iJ_y)^2 + (J_x - iJ_y)^2 = 2(J_x^2 - J_y^2).$$

Hence the Lipkin Hamiltonian can also be written as

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + V(J_x^2 - J_y^2) + \frac{W}{2} J_z^2.$$

Two mapping strategies

There are two main strategies for turning the Lipkin Hamiltonian into Pauli strings:

- 1 **Compressed encoding of the quasispin basis:** encode the $(2j + 1)$ -dimensional quasispin space into $\lceil \log_2(2j + 1) \rceil$ qubits.
- 2 **Direct collective-spin mapping:** write

$$J_\alpha = \frac{1}{2} \sum_i \sigma_i^\alpha$$

on a multi-qubit register and expand the Hamiltonian.

We will use:

- compressed encoding for $N = 4$,
- direct collective mapping for $N = 6$.

Four particles: quasispin dimension and encoding

For $N = 4$,

$$j = \frac{N}{2} = 2, \quad 2j + 1 = 5.$$

Thus the quasispin sector has dimension 5, which fits into 3 qubits since

$$2^2 = 4 < 5 \leq 8 = 2^3.$$

We choose the binary map

$$|2, -2\rangle \leftrightarrow |000\rangle, \quad |2, -1\rangle \leftrightarrow |001\rangle, \quad |2, 0\rangle \leftrightarrow |010\rangle,$$

$$|2, 1\rangle \leftrightarrow |011\rangle, \quad |2, 2\rangle \leftrightarrow |100\rangle.$$

The states $|101\rangle, |110\rangle, |111\rangle$ are unphysical.

Projectors and transition operators on qubits

For one qubit:

$$\begin{aligned} |0\rangle\langle 0| &= \frac{I + Z}{2}, & |1\rangle\langle 1| &= \frac{I - Z}{2}, \\ |1\rangle\langle 0| &= \frac{X - iY}{2}, & |0\rangle\langle 1| &= \frac{X + iY}{2}. \end{aligned}$$

Therefore:

- diagonal operators are expanded with I and Z ,
- off-diagonal transitions are expanded with X and Y ,
- any finite operator in the encoded space can be written as a sum of Pauli strings.

Three-qubit form of J_z

Since

$$J_z = \sum_{m=-2}^2 m |m\rangle \langle m|,$$

we obtain

$$J_z = -2 |000\rangle \langle 000| - |001\rangle \langle 001| + |011\rangle \langle 011| + 2 |100\rangle \langle 100|.$$

Expanding the projectors gives

$$J_z = -\frac{1}{4} IZI + \frac{1}{4} IZZ - \frac{1}{2} ZII - \frac{1}{2} ZIZ - \frac{3}{4} ZZI - \frac{1}{4} ZZZ.$$

Three-qubit form of J_z^2

Similarly,

$$J_z^2 = \sum_{m=-2}^2 m^2 |m\rangle \langle m|,$$

which yields

$$J_z^2 = \frac{5}{4} III + \frac{3}{4} IIZ + IZI + IZZ + \frac{1}{4} ZII - \frac{1}{4} ZIZ.$$

These formulas are exact on the physical 5-dimensional subspace.

Three-qubit form of the ladder terms

For $j = 2$,

$$\begin{aligned} J_+ |-2\rangle &= 2 |-1\rangle, & J_+ |-1\rangle &= \sqrt{6} |0\rangle, \\ J_+ |0\rangle &= \sqrt{6} |1\rangle, & J_+ |1\rangle &= 2 |2\rangle. \end{aligned}$$

Therefore

$$\begin{aligned} J_+^2 + J_-^2 &= \frac{3 + \sqrt{6}}{2} IXI + \frac{\sqrt{6} - 3}{2} IXZ + \frac{\sqrt{6}}{2} XXI + \frac{\sqrt{6}}{2} XXZ \\ &+ \frac{\sqrt{6}}{2} YYI + \frac{\sqrt{6}}{2} YYZ + \frac{3 + \sqrt{6}}{2} ZXI + \frac{\sqrt{6} - 3}{2} ZXZ. \end{aligned}$$

Final $N = 4$, 3-qubit Lipkin Hamiltonian

Substituting into

$$H = \frac{\epsilon}{2}(J_z - 2) + \frac{V}{2}(J_+^2 + J_-^2) + \frac{W}{2}J_z^2$$

gives

$$H_{N=4}^{(3q)} = \left(-\epsilon + \frac{5W}{8}\right) III + \frac{3W}{8} IIZ + \left(-\frac{\epsilon}{8} + \frac{W}{2}\right) IZI + \left(\frac{\epsilon}{8} + \frac{W}{2}\right) IZZ \\ + \left(-\frac{\epsilon}{4} + \frac{W}{8}\right) ZII + \left(-\frac{\epsilon}{4} - \frac{W}{8}\right) ZIZ - \frac{3\epsilon}{8} ZZI - \frac{\epsilon}{8} ZZZ$$

$$+ \frac{V(3 + \sqrt{6})}{4} IXI + \frac{V(\sqrt{6} - 3)}{4} IXZ + \frac{V\sqrt{6}}{4} XXI + \frac{V\sqrt{6}}{4} XXZ + \frac{V\sqrt{6}}{4} YYI + \frac{V\sqrt{6}}{4} YYZ + \frac{V(3 + \sqrt{6})}{4} Z$$

Because the 3-qubit register has 8 basis states but only 5 are physical, one often adds

$$H_{\text{pen}} = \lambda P_{\text{unphys}},$$

where

$$P_{\text{unphys}} = |101\rangle \langle 101| + |110\rangle \langle 110| + |111\rangle \langle 111|.$$

This energetically suppresses leakage outside the encoded quasispin space.

Six particles: direct collective-spin realization

For $N = 6$, a simple and transparent mapping uses six qubits and defines

$$J_x = \frac{1}{2} \sum_{i=1}^6 X_i, \quad J_y = \frac{1}{2} \sum_{i=1}^6 Y_i, \quad J_z = \frac{1}{2} \sum_{i=1}^6 Z_i.$$

This is not qubit-optimal, since the symmetric quasispin sector has dimension 7, but it leads to a very clean Pauli-string Hamiltonian.

Expanding the collective squares

Using $X_i^2 = Y_i^2 = Z_i^2 = I$,

$$J_x^2 = \frac{1}{4} \sum_{i=1}^6 I + \frac{1}{4} \sum_{i \neq j} X_i X_j = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} X_i X_j,$$

and similarly

$$J_y^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Y_i Y_j,$$

$$J_z^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Z_i Z_j.$$

Interaction term on 6 qubits

Using

$$J_+^2 + J_-^2 = 2(J_x^2 - J_y^2),$$

we get

$$\frac{V}{2}(J_+^2 + J_-^2) = V(J_x^2 - J_y^2) = \frac{V}{2} \sum_{i < j} (X_i X_j - Y_i Y_j).$$

Thus the interaction is an all-to-all two-body Pauli interaction.

Linear and quadratic J_z terms on 6 qubits

For the linear term,

$$\frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) = \frac{\epsilon}{2} \left(\frac{1}{2} \sum_{i=1}^6 Z_i - 3I \right) = \frac{\epsilon}{4} \sum_{i=1}^6 Z_i - \frac{3\epsilon}{2} I.$$

For the quadratic term,

$$\frac{W}{2} J_z^2 = \frac{W}{2} \left(\frac{6}{4} I + \frac{1}{2} \sum_{i < j} Z_i Z_j \right) = \frac{3W}{4} I + \frac{W}{4} \sum_{i < j} Z_i Z_j.$$

Final $N = 6$, 6-qubit Lipkin Hamiltonian

Collecting all terms gives

$$H_{N=6}^{(6q)} = \left(-\frac{3\epsilon}{2} + \frac{3W}{4} \right) I + \frac{\epsilon}{4} \sum_{i=1}^6 Z_i + \frac{V}{2} \sum_{i<j} (X_i X_j - Y_i Y_j) + \frac{W}{4} \sum_{i<j} Z_i Z_j.$$

This is already directly usable in qubit-based quantum algorithms.

Comparison of the two Lipkin mappings

Case	Qubits	Main feature
$N = 4$, compressed	3	minimal encoding
$N = 6$, direct	6	simplest Pauli structure

Compressed mappings save qubits but require code-space control. Direct mappings use more qubits but give very transparent Hamiltonians.

Second quantization for fermions

In fermionic many-body theory, the basic operators are

$$a_p^\dagger, \quad a_p,$$

satisfying anticommutation relations

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \quad \{a_p, a_q\} = 0, \quad \{a_p^\dagger, a_q^\dagger\} = 0.$$

A general second-quantized Hamiltonian has the form

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} v_{pqrs} a_p^\dagger a_q^\dagger a_s a_r.$$

Occupation-number basis and qubits

A fermionic Fock basis state is specified by occupations

$$|n_0 n_1 \cdots n_{M-1}\rangle, \quad n_p \in \{0, 1\}.$$

This is already a binary basis, so one may map it directly to M qubits:

$$|n_0 n_1 \cdots n_{M-1}\rangle_{\text{Fock}} \leftrightarrow |n_0 n_1 \cdots n_{M-1}\rangle_{\text{qubit}}.$$

The nontrivial part is not the basis mapping but the operator mapping: fermionic signs must be reproduced correctly on qubits.

Why fermion-to-qubit mappings are needed

A quantum computer acts on qubits, not directly on fermionic Fock states. So we need a mapping

$$a_p^\dagger, a_p, n_p = a_p^\dagger a_p \longrightarrow \text{Pauli strings.}$$

The most common constructions are:

- Jordan–Wigner,
- Bravyi–Kitaev,
- parity mapping.

We focus here on Jordan–Wigner, since it is the most direct mathematically.

Deriving the Jordan–Wigner idea

For a single qubit,

$$\sigma^+ = \frac{X - iY}{2} = |1\rangle\langle 0|, \quad \sigma^- = \frac{X + iY}{2} = |0\rangle\langle 1|.$$

If fermions were distinguishable, we could simply identify

$$a_p^\dagger \sim \sigma_p^+, \quad a_p \sim \sigma_p^-.$$

But this fails because different fermionic modes must anticommute:

$$a_p a_q = -a_q a_p \quad (p \neq q).$$

So we must insert a parity string that counts the number of occupied orbitals before site p .

Jordan–Wigner transformation

For orbital p , the Jordan–Wigner mapping is

$$a_p^\dagger = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p - iY_p}{2}, \quad a_p = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p + iY_p}{2}.$$

The string of Z 's ensures the correct fermionic sign structure. It is often called the *parity string*.

Why the parity string works

Recall that

$$Z |0\rangle = + |0\rangle, \quad Z |1\rangle = - |1\rangle.$$

Therefore

$$\prod_{j=0}^{p-1} Z_j$$

returns a factor

$$(-1)^{n_0+n_1+\dots+n_{p-1}}$$

on a basis state.

This is exactly the fermionic sign picked up when a creation or annihilation operator is moved through all earlier occupied orbitals in the canonical ordering.

Checking the canonical anticommutator

Using the Jordan–Wigner map, one can verify

$$a_p a_p^\dagger + a_p^\dagger a_p = I.$$

Indeed, since the parity string squares to the identity,

$$a_p^\dagger a_p = \frac{1 - Z_p}{2}, \quad a_p a_p^\dagger = \frac{1 + Z_p}{2},$$

and therefore

$$a_p a_p^\dagger + a_p^\dagger a_p = \frac{1 + Z_p}{2} + \frac{1 - Z_p}{2} = I.$$

Number operators under Jordan–Wigner

A key identity is

$$n_p = a_p^\dagger a_p = \frac{1 - Z_p}{2}.$$

Thus one-body diagonal terms become sums of I and Z :

$$\sum_p \epsilon_p a_p^\dagger a_p \longrightarrow \sum_p \epsilon_p \frac{1 - Z_p}{2}.$$

This is one of the most useful simplifications in practical qubit Hamiltonians.

Nearest-neighbor hopping example

For $p < q$, one finds

$$a_p^\dagger a_q + a_q^\dagger a_p$$

maps to

$$\frac{1}{2} (X_p Z_{p+1} \cdots Z_{q-1} X_q + Y_p Z_{p+1} \cdots Z_{q-1} Y_q).$$

So hopping terms become nonlocal Pauli strings because of the fermionic parity chain.

Pair creation operator in full orbital space

Consider a time-reversed pair $(p, \bar{p})$ with $p < \bar{p}$. Then

$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger.$$

Using Jordan–Wigner,

$$a_p^\dagger a_{\bar{p}}^\dagger = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p - iY_p}{2} \left(\prod_{j=0}^{\bar{p}-1} Z_j \right) \frac{X_{\bar{p}} - iY_{\bar{p}}}{2}.$$

After cancelling the common prefix, one gets a Pauli string with a parity chain between p and $\bar{p}$.

Explicit formula for $a_p^\dagger a_q^\dagger$

For $p < q$,

$$a_p^\dagger a_q^\dagger = \frac{1}{4} (X_p - iY_p) Z_{p+1} \cdots Z_{q-1} (X_q - iY_q).$$

Expanding this product gives

$$a_p^\dagger a_q^\dagger = \frac{1}{4} \left(X_p Z \cdots Z X_q - iX_p Z \cdots Z Y_q - iY_p Z \cdots Z X_q - Y_p Z \cdots Z Y_q \right),$$

where the intermediate Z -string runs from $p+1$ to $q-1$.

Hermitian pair transfer operator

A very common Hermitian combination is

$$a_p^\dagger a_q^\dagger + a_q a_p.$$

Under Jordan–Wigner this becomes

$$\frac{1}{2} (X_p Z_{p+1} \cdots Z_{q-1} X_q - Y_p Z_{p+1} \cdots Z_{q-1} Y_q).$$

This is the pairing analogue of the familiar hopping identity, with a sign change between the XX and YY channels.

Local pairing operators

For each time-reversed pair $(p, \bar{p})$, define

$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger, \quad P_p = a_{\bar{p}} a_p,$$

and the pair number operator

$$N_p = a_p^\dagger a_p + a_{\bar{p}}^\dagger a_{\bar{p}}.$$

Now define local quasispin operators

$$S_p^+ = P_p^\dagger, \quad S_p^- = P_p, \quad S_p^z = \frac{1}{2}(N_p - 1).$$

Commutator $[S_p^z, S_p^+]$

We derive the quasispin algebra directly from fermion operators.

Using

$$S_p^z = \frac{1}{2}(N_p - 1), \quad S_p^+ = a_p^\dagger a_{\bar{p}}^\dagger,$$

we compute

$$[S_p^z, S_p^+] = \frac{1}{2}[N_p, a_p^\dagger a_{\bar{p}}^\dagger].$$

Since

$$[N_p, a_p^\dagger] = a_p^\dagger, \quad [N_p, a_{\bar{p}}^\dagger] = a_{\bar{p}}^\dagger,$$

it follows that

$$[N_p, a_p^\dagger a_{\bar{p}}^\dagger] = 2a_p^\dagger a_{\bar{p}}^\dagger.$$

Hence

$$[S_p^z, S_p^+] = S_p^+.$$

Commutator $[S_p^z, S_p^-]$

Similarly,

$$[S_p^z, S_p^-] = \frac{1}{2}[N_p, a_{\bar{p}}a_p].$$

Using

$$[N_p, a_p] = -a_p, \quad [N_p, a_{\bar{p}}] = -a_{\bar{p}},$$

we find

$$[N_p, a_{\bar{p}}a_p] = -2a_{\bar{p}}a_p.$$

Therefore

$$[S_p^z, S_p^-] = -S_p^-.$$

Commutator $[S_p^+, S_p^-]$

Now compute

$$[S_p^+, S_p^-] = a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p - a_{\bar{p}} a_p a_p^\dagger a_{\bar{p}}^\dagger.$$

The first term is

$$a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p = a_p^\dagger n_{\bar{p}} a_p = n_p n_{\bar{p}}.$$

The second term becomes

$$a_{\bar{p}}(1 - n_p)a_{\bar{p}}^\dagger = (1 - n_p)(1 - n_{\bar{p}}).$$

Subtracting yields

$$[S_p^+, S_p^-] = n_p + n_{\bar{p}} - 1 = N_p - 1 = 2S_p^z.$$

Local quasispin algebra

We have proved

$$[S_p^z, S_p^\pm] = \pm S_p^\pm, \quad [S_p^+, S_p^-] = 2S_p^z.$$

So for each pair level p , the operators

$$S_p^+, \quad S_p^-, \quad S_p^z$$

generate an $su(2)$ algebra.

For different levels $p \neq q$, the operators commute:

$$[S_p^\alpha, S_q^\beta] = 0.$$

Thus a pairing model is naturally a sum of commuting local quasispins.

Collective quasispin

Defining

$$S^{\pm} = \sum_p S_p^{\pm}, \quad S^z = \sum_p S_p^z,$$

we immediately get the collective algebra

$$[S^z, S^{\pm}] = \pm S^{\pm}, \quad [S^+, S^-] = 2S^z.$$

This is the precise sense in which the pairing problem can be rewritten as a spin problem. It also explains the close analogy with the Lipkin Hamiltonian.

Nuclear pairing Hamiltonian

A standard reduced pairing Hamiltonian is

$$H_{\text{pair}} = \sum_p \epsilon_p N_p - G \sum_{p,q} P_p^\dagger P_q,$$

where

$$N_p = a_p^\dagger a_p + a_{\bar{p}}^\dagger a_{\bar{p}},$$
$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger, \quad P_p = a_{\bar{p}} a_p.$$

Here p and $\bar{p}$ denote time-reversed partners.

Pairing Hamiltonian in quasispin form

Using

$$S_p^+ = P_p^\dagger, \quad S_p^- = P_p, \quad S_p^z = \frac{1}{2}(N_p - 1),$$

the pairing Hamiltonian becomes

$$H_{\text{pair}} = \sum_p \epsilon_p (2S_p^z + 1) - G \sum_{p,q} S_p^+ S_q^-.$$

Thus the pairing Hamiltonian is an interacting quasispin Hamiltonian. This is the natural starting point for compact qubit encodings.

One pair level as one qubit

For a single pair $(p, \bar{p})$, the paired subspace is two-dimensional:

$$|0\rangle_p \equiv \text{pair unoccupied}, \quad |1\rangle_p \equiv \text{pair occupied}.$$

In this reduced subspace, we may identify

$$S_p^+ = \frac{X_p - iY_p}{2}, \quad S_p^- = \frac{X_p + iY_p}{2}, \quad S_p^z = -\frac{Z_p}{2}.$$

Also,

$$N_p = 1 - Z_p.$$

Pairing Hamiltonian in pair-qubit form

Substituting

$$N_p = 1 - Z_p, \quad P_p^\dagger = \frac{X_p - iY_p}{2}, \quad P_p = \frac{X_p + iY_p}{2},$$

we obtain

$$H_{\text{pair}} = \sum_p \epsilon_p (1 - Z_p) - G \sum_{p,q} S_p^+ S_q^-.$$

Since

$$S_p^+ S_q^- + S_q^+ S_p^- = \frac{1}{2}(X_p X_q + Y_p Y_q),$$

the interaction becomes an XY -type coupling between pair qubits.

Explicit pairing Hamiltonian for Ω pair levels

For Ω pair orbitals, the pairing Hamiltonian in the paired subspace can be written as

$$H_{\text{pair}} = \sum_{p=1}^{\Omega} \epsilon_p (1 - Z_p) - \frac{G}{2} \sum_{p \neq q} (X_p X_q + Y_p Y_q) - G \sum_p S_p^+ S_p^-.$$

Since

$$S_p^+ S_p^- = \frac{1 - Z_p}{2},$$

this yields

$$H_{\text{pair}} = \sum_{p=1}^{\Omega} \left(\epsilon_p - \frac{G}{2} \right) (1 - Z_p) - \frac{G}{2} \sum_{p \neq q} (X_p X_q + Y_p Y_q).$$

Example: two pair levels in pair-qubit space

For two pair orbitals $p = 1, 2$, the paired-subspace Hamiltonian becomes

$$H_{\text{pair}}^{(2)} = \left(\epsilon_1 - \frac{G}{2} \right) (1 - Z_1) + \left(\epsilon_2 - \frac{G}{2} \right) (1 - Z_2) - \frac{G}{2} (X_1 X_2 + Y_1 Y_2).$$

This is a compact two-qubit Hamiltonian ready for variational or time-evolution algorithms.

Full orbital-space example: one pair level

Now keep the two fermionic orbitals p and $\bar{p}$ explicitly as two qubits. The one-level pairing Hamiltonian is

$$H = \epsilon_p(n_p + n_{\bar{p}}) - G a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p.$$

Using Jordan–Wigner,

$$n_p = \frac{1 - Z_p}{2}, \quad n_{\bar{p}} = \frac{1 - Z_{\bar{p}}}{2}.$$

Also

$$a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p = n_p n_{\bar{p}}.$$

Therefore

$$H = \frac{\epsilon_p}{2}(2 - Z_p - Z_{\bar{p}}) - \frac{G}{4}(1 - Z_p)(1 - Z_{\bar{p}}).$$

Expanded one-pair full-orbital Pauli form

Expanding the previous result gives

$$H = \left(\epsilon_p - \frac{G}{4} \right) I + \left(-\frac{\epsilon_p}{2} + \frac{G}{4} \right) Z_p + \left(-\frac{\epsilon_p}{2} + \frac{G}{4} \right) Z_{\bar{p}} - \frac{G}{4} Z_p Z_{\bar{p}}.$$

So even the full-orbital one-pair Hamiltonian reduces to a simple two-qubit diagonal Pauli Hamiltonian.

Full orbital-space example: pair transfer between two levels

Consider two pair levels p and q , with orbitals

$$(p, \bar{p}, q, \bar{q}).$$

A pairing transfer term is

$$P_p^\dagger P_q + P_q^\dagger P_p = a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{q}} a_q + a_q^\dagger a_{\bar{q}}^\dagger a_{\bar{p}} a_p.$$

Under Jordan–Wigner, this becomes a Hermitian sum of four-qubit Pauli strings.

Explicit Pauli decomposition of pair transfer

Assume the orbital ordering

$$p < \bar{p} < q < \bar{q}.$$

Then one finds

$$\begin{aligned} P_p^\dagger P_q + P_q^\dagger P_p = & \frac{1}{8} \left(X_p X_{\bar{p}} X_q X_{\bar{q}} - X_p Y_{\bar{p}} Y_q X_{\bar{q}} \right. \\ & \left. - Y_p X_{\bar{p}} X_q Y_{\bar{q}} + Y_p Y_{\bar{p}} Y_q Y_{\bar{q}} \right) \end{aligned}$$

up to the parity-string conventions induced by the chosen ordering. In adjacent pair ordering, the intermediate Z -chains cancel within the paired sector, leaving a compact four-body expression.

Interpretation of the full-orbital pairing terms

This explicit decomposition shows:

- full fermionic orbital space leads to longer Pauli strings,
- pair-qubit reduction produces shorter and more structured qubit Hamiltonians,
- both are exact, but they operate in different Hilbert spaces.

So there is a practical tradeoff:

- full-space mappings are fully general,
- reduced quasispin/pair encodings are more compact when the physics is dominated by seniority-zero pairing.

Lipkin versus pairing: structural comparison

Both Hamiltonians admit spin-like descriptions:

- Lipkin:

$$H_{\text{Lipkin}} = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2,$$

- Pairing:

$$H_{\text{pair}} = \sum_p \epsilon_p N_p - G \sum_{p,q} P_p^\dagger P_q.$$

Both become sums of Pauli strings after:

- choosing an encoding,
- expressing collective or pair operators in terms of X, Y, Z .

General recipe: from second quantization to qubits

For a fermionic Hamiltonian:

- 1 write the Hamiltonian in terms of $a_p^\dagger, a_p$,
- 2 decide whether a reduced quasispin or pair-qubit subspace is appropriate,
- 3 otherwise apply Jordan–Wigner or a related mapping,
- 4 simplify number and pair operators,
- 5 collect the resulting Pauli strings.

For collective-spin models:

- 1 write the model in J_x, J_y, J_z or ladder-operator form,
- 2 choose compressed or direct qubit encoding,
- 3 expand diagonal and off-diagonal matrix elements into Pauli strings.

Summary

In this expanded version we added:

- a fuller Jordan–Wigner derivation,
- a formal derivation of the quasispin algebra directly from second quantization,
- explicit Pauli decompositions for small pairing examples in full orbital space,
- and a more systematic comparison of compact versus full-space encodings.

The unifying message is:

fermions or collective spins $\longrightarrow$ spin algebra $\longrightarrow$ Pauli strings.

Quantum Fourier transform: Overarching motivation

We will start with a reminder on Fourier theory and in particular on discrete Fourier transforms (DFTs). There are many motivations for the DFTs. For those of you familiar with signal processing, harmonic oscillations, and many other areas of applications, Fourier transforms are almost standard kitchen items.

But first some general motivation for Quantum Fourier Transforms (QFTs).

Quantum Fourier Transforms (QFTs)

- ① QFTs are the quantum analogue of the Discrete Fourier Transforms (DFTs).
- ② They play a crucial role in quantum algorithms like Shor's Algorithm and Quantum Phase Estimation.
- ③ QFTs provide an *exponential speedup* over classical Fourier Transform methods.

Why Quantum Fourier Transforms and exponential speedup

- 1 Classical Discrete Fourier Transforms (DFTs) require $O(N^2)$ operations.
- 2 The Fast Fourier Transform (FFT) improves this to $O(N \log N)$ operations.
- 3 QFTs reduce complexity to $O((\log N)^2)$ using quantum gates.

Quantum Fourier Transforms and quantum parallelism

- ① QFTs act on a *superposition* of states, processing all inputs simultaneously.
- ② This is crucial in algorithms like:
 - ① Shor's Algorithm for factoring large numbers.
 - ② Quantum Phase Estimation (QPE) for eigenvalue extraction.

Quantum Fourier Transforms and implementations

- ① QFTs require only *Hadamard gates and controlled-phase gates*.
- ② A 3-qubit QFT circuit uses only $O(n^2)$.
- ③ This makes QFTs highly efficient for *quantum hardware*.

Quantum Fourier Transform and Applications in Quantum Computing

- ① **Shor's Algorithm:** Uses QFTs to find periodicity in modular exponentiation.
- ② **Quantum Phase Estimation (QPE):** QFTs extract eigenvalues of unitary matrices.
- ③ **Quantum Signal Processing:** QFTs enable spectral analysis on quantum data.
Important for AI applications.

Quantum Fourier Transforms and Reduced Memory Complexity (not covered)

- 1 Classical DFTs require $O(N)$ space.
- 2 QFT store Fourier-transformed coefficients *implicitly* in qubit amplitudes.
- 3 Only $O(n)$ qubits are needed for a size $N = 2^n$ transformation.

Quantum Fourier Transforms and Quantum Signal Processing (not covered in this course)

QFTs enable applications such as:

- 1 Quantum spectral analysis.
- 2 Quantum image processing.
- 3 Quantum filtering and denoising.

These can enhance AI, cryptography, and data processing.

Why Quantum Fourier Transforms? Brief summary

- 1 Quantum Fourier Transforms provide *exponential speedup*
- 2 They enable key quantum algorithms like *Shor's Algorithm* and *Quantum Phase Estimation*.
- 3 QFTs are more efficient in *memory usage and circuit complexity* than classical methods.
- 4 Future quantum applications in *signal processing, AI, and cryptography* will heavily rely on QFT.